

Detailed Cybersecurity Lab Reports Compilation

This document contains detailed reports generated from multiple cybersecurity lab activities including Alert Management, Alert Triage, Evidence Preservation, Incident Response, and Capstone Project execution using industry-standard cybersecurity tools.

1. Alert Management Practice

The objective of this activity was to understand the process of monitoring, analyzing, and prioritizing security alerts using the Wazuh SIEM platform. The Threat Hunting module was used to review generated alerts, examine alert severity levels, and understand MITRE ATT&CK mappings associated with suspicious events. During the exercise, multiple security alerts related to authentication activity, system monitoring, and configuration changes were reviewed. The lab demonstrated how Security Operations Center (SOC) analysts monitor security events, investigate suspicious behavior, and maintain documentation during daily monitoring operations.

Main Points:

- Wazuh dashboard configured successfully
- Threat Hunting reports reviewed
- Alert severity analysis performed
- MITRE ATT&CK mappings analyzed
- Practical SOC monitoring workflow understood

2. Alert Triage Practice

This activity focused on performing alert triage for a simulated brute-force SSH attack using Wazuh, AlienVault OTX, and VirusTotal. Security events were reviewed inside the Wazuh Threat Hunting module to identify suspicious login attempts and authentication failures. Raw logs were filtered using KQL queries, and suspicious IP addresses were analyzed using threat intelligence platforms. The investigation process included documenting findings, examining timestamps, identifying repeated authentication attempts, and determining whether the activity represented a false positive or an actual threat.

Main Points:

- Brute-force SSH activity investigated
- Threat Hunting events reviewed
- Threat intelligence validation performed
- Alert escalation process documented
- SOC triage workflow demonstrated

3. Evidence Preservation and Collection

This lab demonstrated evidence preservation and forensic acquisition techniques using Velociraptor, FTK Imager, and SHA256 hashing. Volatile network evidence was collected

using Velociraptor queries such as `SELECT * FROM netstat` to identify active connections and processes. Memory acquisition was performed to capture volatile artifacts including running processes, network sessions, and malware traces. FTK Imager was used for memory capture operations and forensic evidence handling. SHA256 hashing was performed to verify evidence integrity and ensure no modifications occurred during acquisition and transfer.

Main Points:

- Volatile data collection performed
- Memory acquisition completed
- Evidence hashing verified integrity
- Chain-of-custody documentation maintained
- Forensic evidence handling procedures followed

4. Incident Response Documentation

A simulated cybersecurity investigation and incident response process was conducted using Kali Linux and Metasploitable2 in a controlled virtual lab environment. Vulnerability assessment was performed using Nmap to identify exposed services including FTP, SSH, Telnet, Apache HTTP Server, and MySQL. Service enumeration, process investigation, and system log analysis were conducted to understand system behavior and identify suspicious activities. Controlled exploitation testing using Metasploit Framework was performed to validate known vulnerabilities and simulate real-world attack scenarios.

Main Points:

- Network connectivity verified
- Vulnerability scanning performed using Nmap
- Service enumeration completed
- Log analysis conducted
- Metasploit exploitation demonstrated
- Incident response procedures documented

5. Capstone Project: Full Alert-to-Response Cycle

The capstone project demonstrated a complete cybersecurity workflow involving attack simulation, detection, triage, incident response, and reporting. Metasploit Framework was used to exploit the VSFTPD vulnerable FTP service running on Metasploitable2. Wazuh monitored system events and provided security visibility into suspicious activities occurring during the attack lifecycle. CrowdSec was configured to create blocking decisions against the attacker IP address to simulate containment and response procedures. The exercise highlighted the importance of integrating offensive and defensive security tools to improve SOC visibility and incident response capabilities.

Main Points:

- VSFTPD exploit successfully executed
- Meterpreter session established

- Wazuh monitoring activity analyzed
- CrowdSec blocking decision added
- Alert-to-response workflow demonstrated
- Practical SOC operations experience gained

Executive Summary

The cybersecurity lab exercises successfully demonstrated practical concepts related to Security Operations Center (SOC) monitoring, vulnerability assessment, alert triage, evidence preservation, forensic analysis, and incident response procedures.

Multiple industry-standard security tools including Wazuh, Metasploit Framework, CrowdSec, Velociraptor, FTK Imager, AlienVault OTX, VirusTotal, and Nmap were utilized throughout the labs. These activities provided practical exposure to attack simulation, security monitoring, evidence collection, threat intelligence analysis, and response workflows commonly used in modern cybersecurity environments.

Recommendations

- **Enable active remediation and bouncer components in CrowdSec for automated IP blocking.**
- **Improve SIEM correlation rules and alert prioritization mechanisms.**
- **Regularly patch vulnerable systems and disable unnecessary services.**
- **Perform continuous security monitoring and vulnerability assessments.**
- **Maintain proper forensic evidence handling and chain-of-custody procedures.**
- **Use multiple threat intelligence platforms during investigations.**
- **Conduct regular penetration testing exercises to identify security weaknesses.**
- **Implement stronger authentication and access control mechanisms.**

Conclusion

The combined cybersecurity lab exercises provided valuable hands-on experience in offensive security testing, defensive monitoring, forensic evidence preservation, and incident response workflows. The projects demonstrated how cybersecurity tools work together to detect, investigate, and respond to security incidents within controlled virtual environments. These activities improved practical knowledge of SOC operations, threat detection, vulnerability management, forensic analysis, and incident handling methodologies used by cybersecurity professionals.